RACS — Complete Project Overview

Retrieval-Augmented Coverless Semantic Steganography

> ***Purpose of this document:*** *This is the final project-level technical overview before implementation planning. It establishes the existing AgentSemSteCom system, its reported results, the problem we are targeting, the finalized RACS architecture, the role of FAISS and VLM/agentic reranking, research questions, metrics, comparison strategy, risks, and the intended novelty.*

> *The next document should turn this overview into an implementation plan without changing the experimental comparison framework.*

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1. Executive Summary

We are building a research system called **RACS (Retrieval-Augmented Coverless Semantic Steganography)**.

The starting point is the 2026 IEEE paper:

“Secure Intellicise Wireless Network: Agentic AI for Coverless Semantic Steganography Communication”

The paper proposes **AgentSemSteCom**, a coverless semantic steganography system that uses agentic AI, semantic extraction, implicit structural features, diffusion-based reference-image generation, EDICT-based reversible diffusion, digital-token security, and semantic communication.

The key limitation we want to investigate is the cost of the **reference-image generation stage**.

In the original system, a reference image is generated through a diffusion process controlled by the public semantic key, implicit structural feature, and digital token. This creates a computationally expensive reference-acquisition step.

Our proposed idea is:

> ***Instead of always generating the reference image from scratch, retrieve suitable real images from a pre-indexed image database and use intelligent multimodal reasoning to select the most compatible candidate.***

The finalized RACS idea therefore has three levels:

```
Level 1:
FAISS retrieval
        ↓
Fast candidate discovery

Level 2:
Structure-aware + VLM/agentic reranking
        ↓
Intelligent candidate selection

Level 3:
Optional lightweight editing
        ↓
Improve structural compatibility when necessary
```

The important design principle is:

> ***The VLM/agent should make retrieval smarter, not replace retrieval.***

We therefore do **not** put an expensive or nondeterministic LLM directly in place of FAISS.

Instead:

```
FAISS
  ↓
Top-K candidates
  ↓
Structural filtering
  ↓
Deterministic/shared VLM reranking
  ↓
High-quality candidate pool
  ↓
Digital-token-controlled selection
  ↓
Xref
```

The rest of the coverless steganography and semantic communication pipeline remains as close as possible to AgentSemSteCom.

The core research question is:

> ***Can intelligent retrieval-based reference-image acquisition replace expensive diffusion-based reference-image generation while maintaining acceptable reconstruction quality, security, covertness, and channel robustness, while substantially reducing latency and computational cost?***

We are **not assuming retrieval will automatically be better**.

The project is an experimental comparison. Retrieval may:

- significantly reduce latency,
- reduce GPU computation,
- maintain comparable quality,
- or introduce quality degradation because the retrieved image is not sufficiently compatible with the downstream EDICT process.

All of these outcomes are scientifically useful.

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2. Existing Solution: AgentSemSteCom

2.1 What problem does the original paper solve?

Traditional semantic steganography may require a cover image or depend strongly on semantic keys.

The AgentSemSteCom paper attempts to remove the need for an externally supplied cover image and strengthen security against intelligent eavesdroppers.

The system creates a **coverless stego image** by:

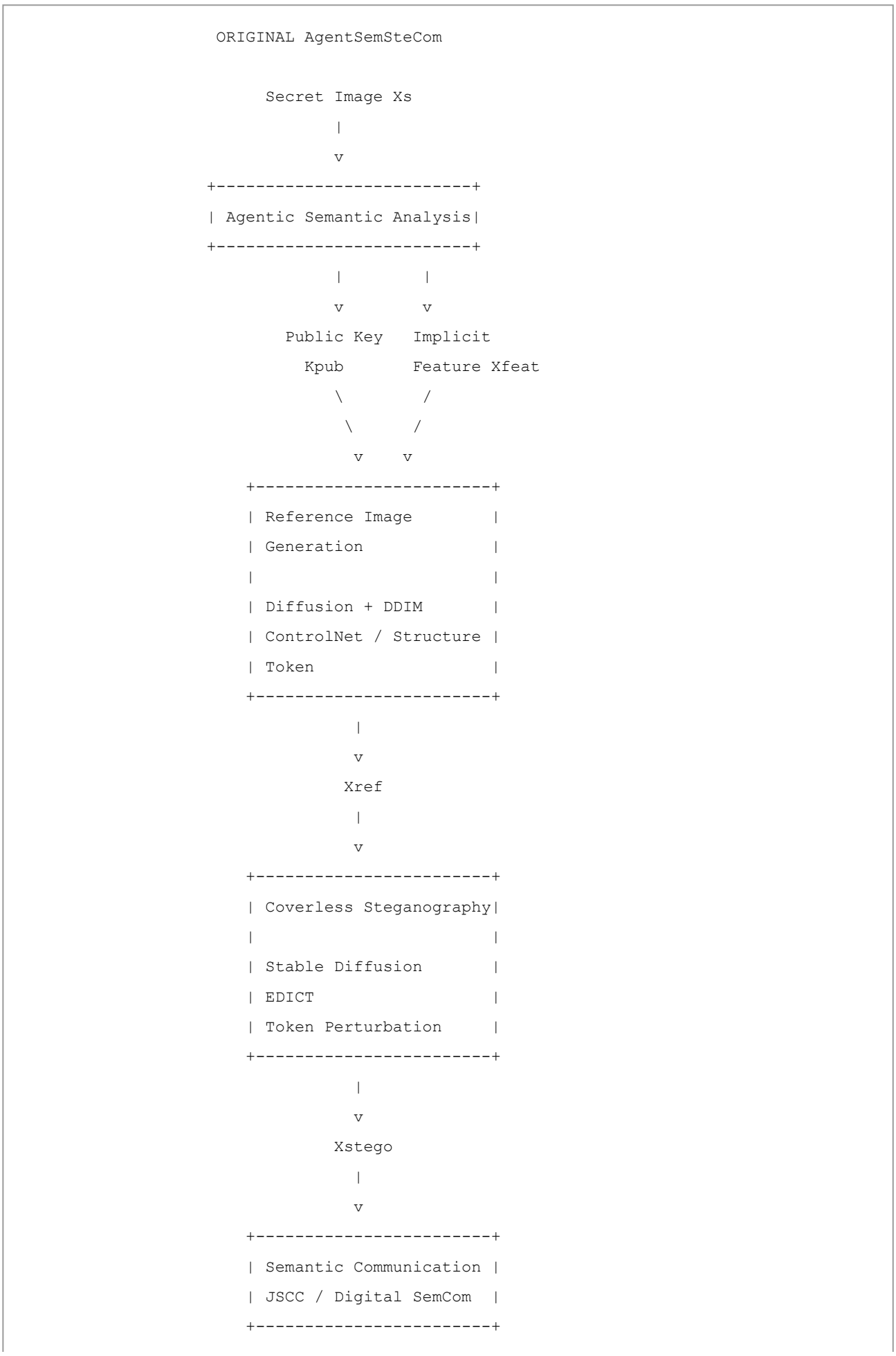
1. extracting semantic and structural information from the secret image,
2. generating a reference image,
3. using an invertible diffusion mechanism to create the stego image,
4. transmitting it through semantic communication,
5. reconstructing the secret at the receiver.

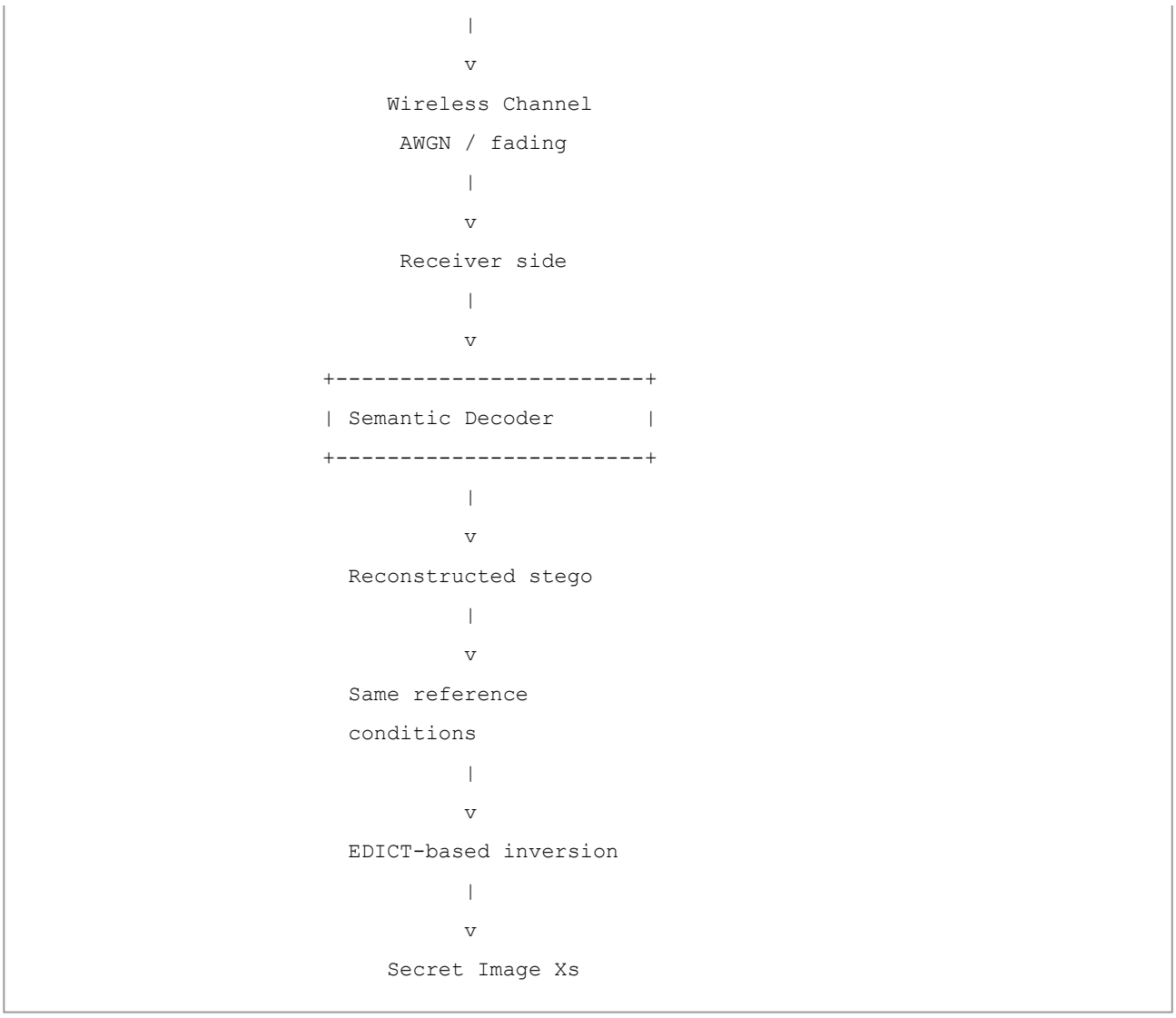
The paper introduces a **digital token** as an additional private control mechanism.

The token participates in both reference generation and latent perturbation, creating a dual-stage protection mechanism.

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3. Existing System — High-Level Pipeline


ORIGINAL AgentSemSteCom
Secret Image Xs
Agentic Semantic Analysis
Public Key
Kpub
Implicit
Feature Xfeat
Reference Image
Generation
Diffusion + DDIM
ControlNet / Structure
Token
Xref
Coverless Steganography
Stable Diffusion
EDICT
Token Perturbation
Xstego
Semantic Communication
JSCC / Digital SemCom



The existing system's six major stages are the foundation for our comparison.

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4. Existing System — Six Main Stages

The original paper organizes the system into six major stages.

Stage 1 — Semantic Extraction and Public Key Generation

The secret image is analyzed by the agentic AI system.

Two important outputs are produced.

Public semantic key

The system generates:

```
Kpub = fLLM(Xs)
```

The public key describes the semantic content of the secret image.

Example:

```
Secret image
     ↓
LLM / semantic model
     ↓
"a wild boar walking through a lush green field"
```

The public key therefore tells the system **what the image is about**.

Implicit structural feature

The system also extracts:

```
Xfeat = G(Xs, R)
```

The feature depends on the task/class.

Examples include:

- semantic segmentation for common steganography images,
- OpenPose skeletal landmarks for facial images,
- style-related conditions for style images.

The important distinction is:

```
Kpub  → semantic/content information

Xfeat → structural/layout information
```

This distinction becomes especially important for RACS because semantic similarity alone may not guarantee structural compatibility.

---

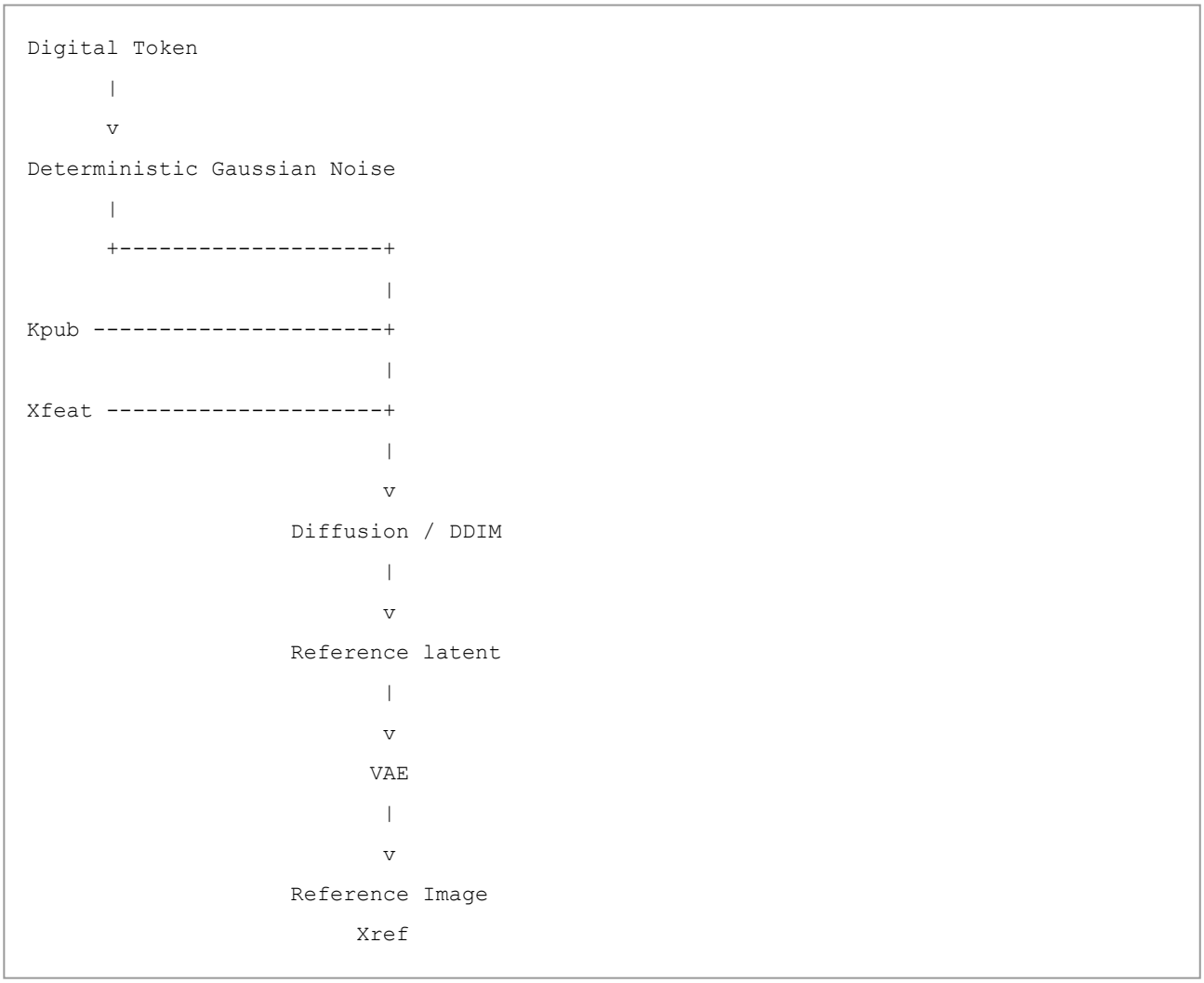
5. Stage 2 — Original Reference Image Generation

This is the main part of the system that our research changes.

The original system uses diffusion-based generation.

The digital token is used to initialize the diffusion process deterministically.

Conceptually:



ControlNet/structural conditioning injects structural information into the generation process.

The result is a reference image that is semantically and structurally conditioned on the secret image.

This stage is expensive because it requires repeated diffusion sampling.

---

6. Stage 3 — Coverless Steganography

After the reference image is obtained, the secret image is processed through the coverless steganography mechanism.

The paper uses:

- Stable Diffusion v1.5
- EDICT
- coupled latent trajectories
- latent perturbation controlled by the digital token
- VAE

The important idea behind EDICT is reversible/invertible diffusion.

Instead of simply generating a visually similar image, the system attempts to preserve enough information for the original secret image to be recovered.

Conceptually:

```
Secret image
     |
     v
VAE encoder
     |
     v
Secret latent Zs
     |
     v
Forward diffusion
     |
     v
Noisy latent ZT
     |
     v
Token-controlled latent perturbation
     |
     v
EDICT reverse diffusion
conditioned by Kpub + Xref
     |
     v
VAE decoder
     |
     v
Stego image
```

---

7. Digital Token — Two Roles in the Existing System

The digital token is important because it is not used for only one purpose.

Role 1 — Reference generation

The token acts as a deterministic random seed.

```
Token + semantic conditions
        ↓
Reference-generation trajectory
```

Role 2 — Latent perturbation

The token also determines the latent perturbation used during the coverless steganography process.

Therefore:

```
Digital Token
      |
      +----> Reference generation seed
      |
      +----> Latent perturbation control
```

This provides the existing system with dual-stage protection.

---

8. Stage 4 — Semantic Communication

The generated stego image is encoded for transmission.

The paper evaluates both digital and analog semantic communication settings.

For analog semantic communication:

- SwinJSCC
- AWGN channel
- SNR values from 5 to 20 dB
- channel coefficient h = 1
- channel bandwidth C = 48
- trained SNR = 10 dB

For digital semantic communication:

- MDJCM
- DIV2K training setup
- modulation order 64
- SNR = 10 dB

For our main comparison, the communication configuration should remain unchanged.

---

9. Stage 5 — Receiver Recovery

The receiver obtains the reconstructed stego image after semantic communication.

It then performs the reverse process.

```
Received stego image
        |
        v
Semantic decoder
        |
        v
Reconstructed stego image
        |
        v
Reference / semantic conditions
        |
        v
EDICT inversion
        |
        v
Inverse token perturbation
        |
        v
Recovered secret image
```

The correct digital token is required for legitimate recovery.

---

10. Stage 6 — Optional Task-Oriented Enhancement

The original paper also demonstrates an optional enhancement module.

For facial images, it uses **CodeFormer** to improve texture/detail.

This is not the fundamental hiding/recovery mechanism.

It is an optional receiver-side enhancement step and should not be mixed into the core RACS comparison unless explicitly being evaluated.

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11. Existing System — Main Implementation Parameters

The reported practical setup includes:

Component	Reported setup
Diffusion backbone	Stable Diffusion v1.5
EDICT steps	50
EDICT perturbation strength η	0.05
EDICT mixing coefficient	0.93
Edit strength	0.5
Reference guidance	IPAdapter-plus
IPAdapter weight	1
Analog semantic communication	SwinJSCC
Digital semantic communication	MDJCM
Analog channel	AWGN
SNR range	5, 10, 15, 20 dB
Channel coefficient	h = 1
Analog channel bandwidth	C = 48
Digital modulation order	64
Facial enhancement	CodeFormer
CodeFormer fidelity weight	0.5

These values should form the baseline configuration for our reproduction wherever feasible.

---

12. Existing Evaluation Metrics

The paper uses four primary image-reconstruction metrics.

12.1 PSNR — Peak Signal-to-Noise Ratio

Measures pixel-level reconstruction quality.

Higher is better.

```
Higher PSNR
      ↓
Recovered image is closer to original
```

12.2 MSE — Mean Squared Error

Measures average squared pixel difference.

Lower is better.

```
Lower MSE
      ↓
Smaller reconstruction error
```

The paper reports MSE scaled by `10^-2` in its table.

12.3 SSIM — Structural Similarity Index

Measures structural similarity.

Higher is better.

```
Higher SSIM
      ↓
Structural organization is better preserved
```

12.4 LPIPS — Learned Perceptual Image Patch Similarity

Measures perceptual similarity using deep features.

Lower is better.

```
Lower LPIPS
      ↓
Perceptually more similar
```

---

13. Existing System — Reported Results

The original paper evaluates AgentSemSteCom on the **UniStega dataset** and tests different image classes and SNR conditions.

The paper reports:

- common steganography images,
- facial images,
- style images,

at:

```
SNR = 5, 10, 15, 20 dB
```

The paper states that AgentSemSteCom generally maintains:

- PSNR above 20 dB for almost all tested cases,
- MSE on the order of `10^-2`.

The facial class generally performs best, which the paper attributes to the OpenPose skeletal-landmark structural feature.

---

14. Important Baseline Numbers From the Existing Paper

These are the numbers we should preserve as reference points.

14.1 General AgentSemSteCom performance

At SNR = 5 dB for the facial class:

```
PSNR = 22.73 dB
SSIM = 0.6989
```

For the style class at SNR = 5 dB:

```
MSE   = 0.721 × 10^-2
LPIPS = 0.3850
```

At SNR = 20 dB, the facial class reaches:

```
PSNR = 26.29 dB
LPIPS = 0.2127
```

---

15. AgentSemSteCom vs SemSteDiff

For the common steganography class at SNR = 5 dB:

Metric	AgentSemSteCom	SemSteDiff
PSNR	19.71 dB	18.04 dB
MSE	1.207 × 10^-2	1.73 × 10^-2

At SNR = 10 dB:

Metric	AgentSemSteCom	SemSteDiff
SSIM	0.5643	0.5183

These values form an additional external baseline.

---

16. Optional Enhancement Results in the Existing Paper

For facial images at SNR = 5 dB:

Metric	Before enhancement	After enhancement
PSNR	22.74 dB	23.15 dB
SSIM	0.6989	0.7244

Metric	Before enhancement	After enhancement
MSE	0.566 × 10^-2	0.514 × 10^-2
LPIPS	0.4042	0.2928

At SNR = 20 dB:

```
Enhanced PSNR = 25.59 dB
Baseline PSNR = 26.29 dB
```

This is important because perceptual enhancement can improve LPIPS while not necessarily improving pixel-level PSNR.

Therefore our evaluation must never depend on a single metric.

---

17. Existing Security Result

The paper compares a legitimate receiver using the correct digital token against an eavesdropper using a wrong token.

Reported result:

```
Correct token:
PSNR = 25.37 dB

Wrong token:
PSNR = 18.91 dB
```

Difference:

```
25.37 - 18.91 = 6.46 dB
```

This provides an important baseline for our security experiments.

RACS should reproduce an equivalent conceptual security experiment.

---

18. What Is Missing From the Existing Evaluation?

The original paper focuses heavily on:

- reconstruction quality,

- semantic communication robustness,
- security,
- image classes,
- SNR.

Our project adds a major efficiency dimension.

```
Quality
   +
Security
   +
Robustness
   +
Latency
   +
Computational cost
   +
Memory usage
```

This is important because the research motivation is not simply to obtain another image-generation method.

The objective is to determine whether a different reference-acquisition strategy can make the system more practical.

---

19. The Problem We Are Targeting

The original reference-image generation pipeline is computationally expensive.

Conceptually:

```
Secret image
     ↓
Kpub + Xfeat
     ↓
Diffusion initialization
     ↓
Multiple denoising / sampling steps
     ↓
Reference image
```

The reference image is generated even though a large database may already contain real images with similar semantic content.

This motivates:

> ***Can we replace expensive generation with retrieval?***

Instead of:

```
Generate an image
```

we perform:

```
Search a database
     ↓
Retrieve suitable candidates
     ↓
Intelligently select the best candidate
     ↓
Optionally perform lightweight editing
```

---

20. Proposed System — RACS

20.1 Core idea

RACS stands for:

Retrieval-Augmented Coverless Semantic Steganography

The core modification is:

```
Original:

Kpub + Xfeat + Token
          ↓
Diffusion generation
          ↓
Reference image



RACS:

Kpub + Xfeat
          ↓
FAISS retrieval
          ↓
Top-K candidates
          ↓
Structure-aware filtering
          ↓
VLM / agentic reranking
          ↓
Token-controlled deterministic selection
          ↓
Optional lightweight editing
          ↓
Reference image
```

The remainder of the steganographic pipeline should initially remain as close as possible to the original system.

---

21. Why We Should Not Simply Call This “RAG”

The project uses retrieval-augmented ideas, but standard RAG normally means:

```
Query
  ↓
Retrieve documents
  ↓
Give documents to an LLM
  ↓
Generate an answer
```

Our system is different.

We are doing:

```
Semantic / structural query
        ↓
Image retrieval
        ↓
Candidate evaluation
        ↓
Reference image
```

A more technically precise description is:

- Retrieval-Augmented Reference Image Selection
- Multimodal Retrieval-Based Reference Conditioning
- Retrieval-Based Reference Image Acquisition

“RAG” may be used informally, but the research description should clearly define that retrieval is operating over **images**, not text documents.

---

22. Why Not Use an LLM/VLM Instead of Simple Retrieval?

This is an important design decision.

A simple retrieval system can do:

```
Query
 ↓
FAISS
 ↓
Highest similarity image
```

This is fast, but semantic similarity alone may not be enough.

A VLM/agent can evaluate multiple retrieved candidates for:

- semantic compatibility,
- structural compatibility,
- spatial arrangement,
- visual context,
- texture,
- suitability for downstream steganography.

Therefore the strongest design is not:

```
LLM instead of FAISS
```

but:

```
FAISS
  ↓
Top-K candidates
  ↓
VLM/agent evaluates candidates
  ↓
Best candidate
```

This gives us the advantages of both approaches.

---

23. Arguments for Simple Retrieval

Advantages

1. Very low online computation

The database embeddings are precomputed.

Runtime mainly performs:

```
Query embedding
+
FAISS search
```

2. Very low latency

Nearest-neighbor retrieval can be substantially cheaper than diffusion generation.

3. Reproducibility

The same:

- query,
- database,
- embedding model,
- index,
- preprocessing

can produce the same candidate.

4. Clean research comparison

Replacing only diffusion reference generation with FAISS gives a very clear experiment:

```
Generation → Retrieval
```

5. Strong efficiency story

If the quality remains acceptable, the computational reduction becomes easy to demonstrate.

---

24. Arguments Against Simple Retrieval

The main weakness is that nearest-neighbor semantic similarity does not guarantee structural compatibility.

Example:

```
Secret:

person standing next to a car

Retrieved:

person sitting inside a car
```

The semantic meaning is similar.

The spatial structure is not.

Because the downstream EDICT process uses the reference image as conditioning, this mismatch may reduce recovery quality.

Therefore:

> ***Pure CLIP similarity may be too weak for the final system.***

This motivates intelligent reranking.

---

25. Arguments for VLM/Agentic Reranking

A VLM can inspect the Top-K retrieved images and evaluate them more intelligently.

For example:

```
Candidate 1:
Semantic match = high
Structure match = medium

Candidate 2:
Semantic match = high
Structure match = high

Candidate 3:
Semantic match = medium
Structure match = very high
```

A VLM/agent can select the candidate with the best overall compatibility.

This provides:

- better semantic understanding,
- structural reasoning,
- candidate comparison,
- stronger research novelty,
- closer alignment with the agentic-AI philosophy of the original paper.

---

26. Arguments Against VLM/Agentic Reranking

The main problem is computational cost.

If we retrieve 100 candidates and run a large VLM on all of them, the efficiency advantage can shrink.

There is also a reproducibility problem if the VLM is:

- cloud-based,
- nondeterministic,
- temperature-dependent,
- changing between versions.

Therefore we should **not** use an arbitrary online LLM API inside the critical reference-selection path.

The preferred design is:

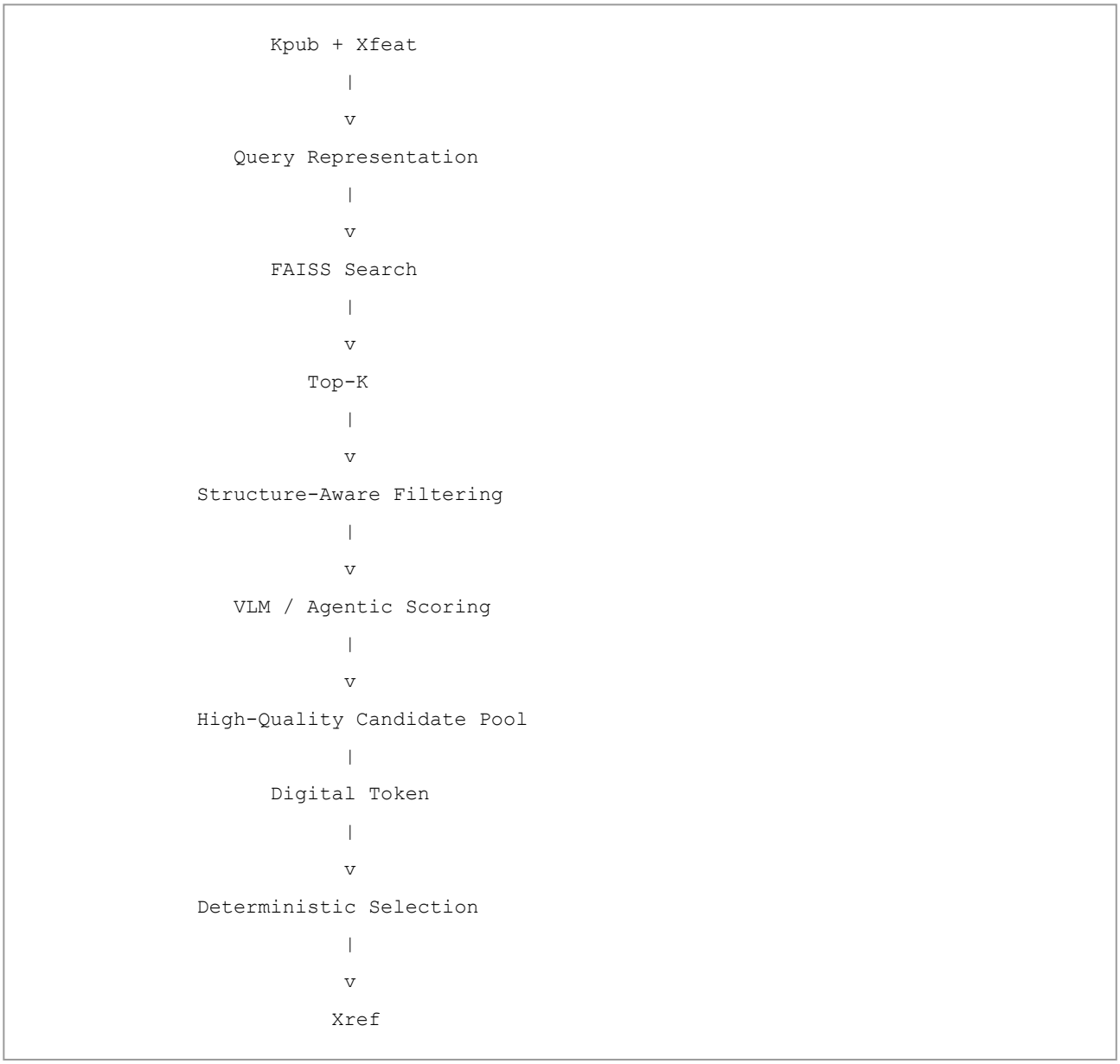
```
FAISS
 ↓
Top-K
 ↓
Light structural filtering
 ↓
Small/frozen/shared VLM
 ↓
Deterministic score
 ↓
Token selection
```

The same model and configuration are used at both transmitter and receiver.

---

27. Final Retrieval Architecture

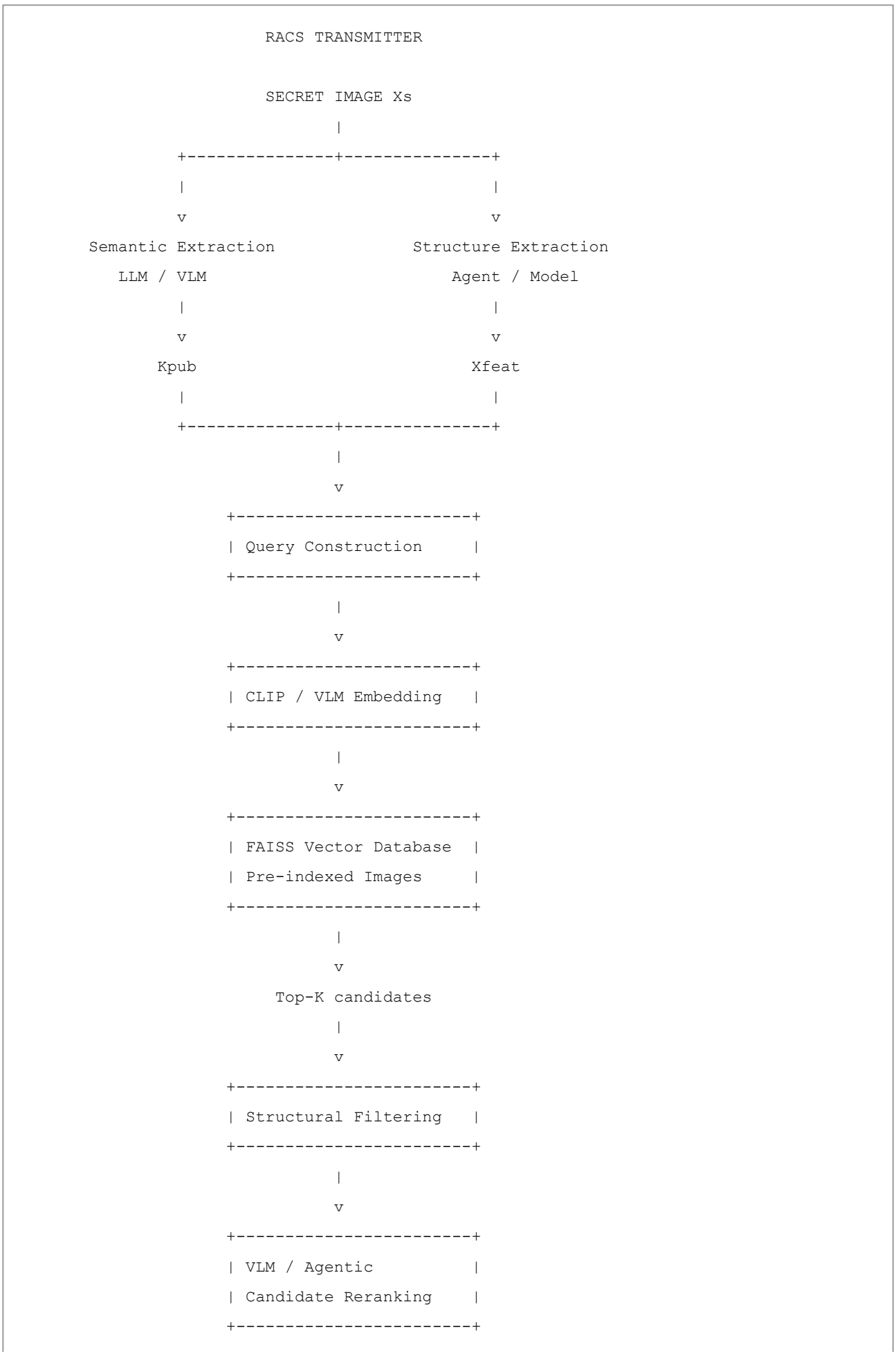
The finalized RACS reference acquisition module is:

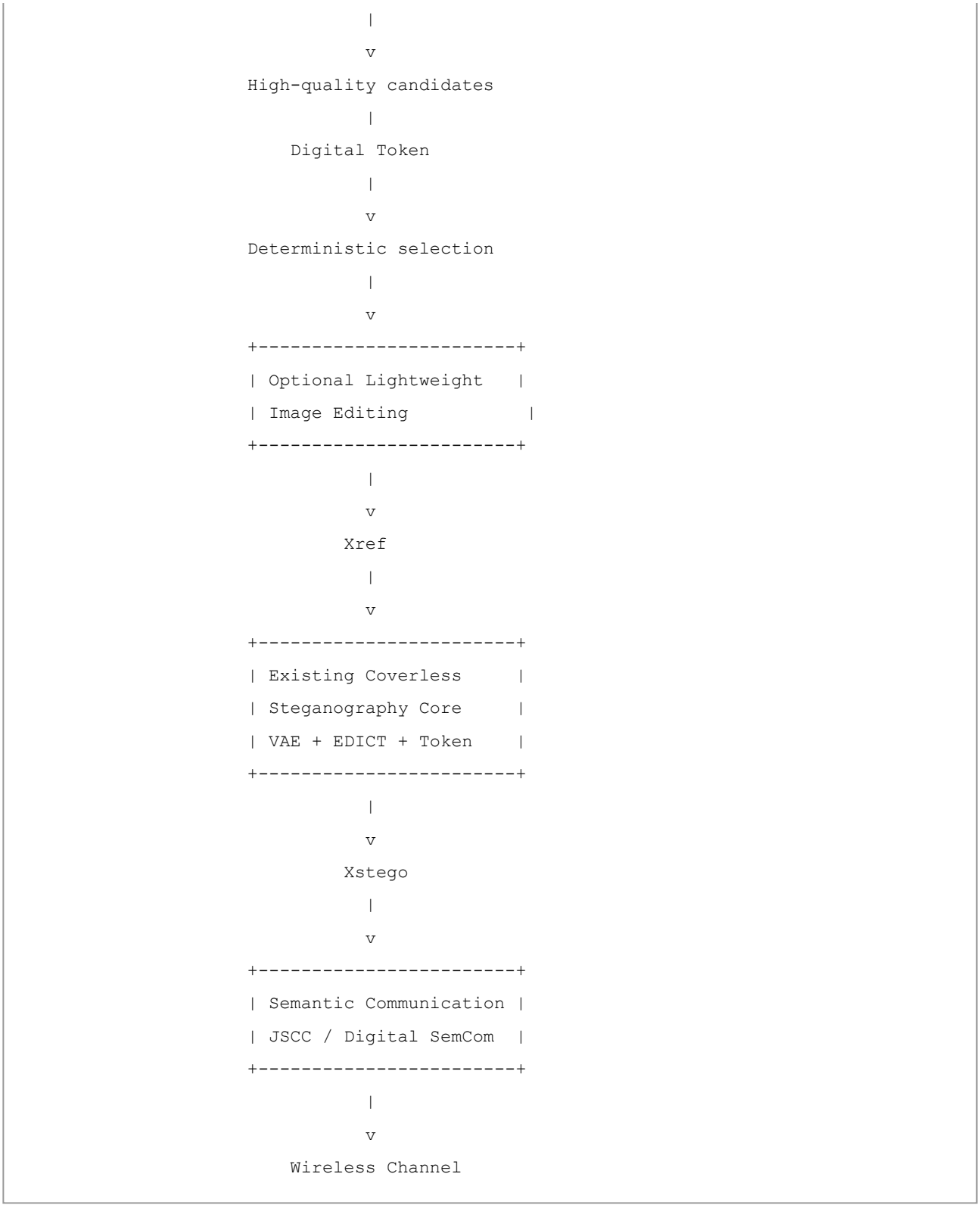


This is the main architecture we should implement and evaluate.

---

28. Proposed Architecture


RACS TRANSMITTER
SECRET IMAGE Xs
Semantic Extraction
LLM / VLM
Kpub
Structure Extraction
Agent / Model
Xfeat
Query Construction
CLIP / VLM Embedding
FAISS Vector Database
Pre-indexed Images
Top-K candidates
Structural Filtering
VLM / Agentic
Candidate Reranking

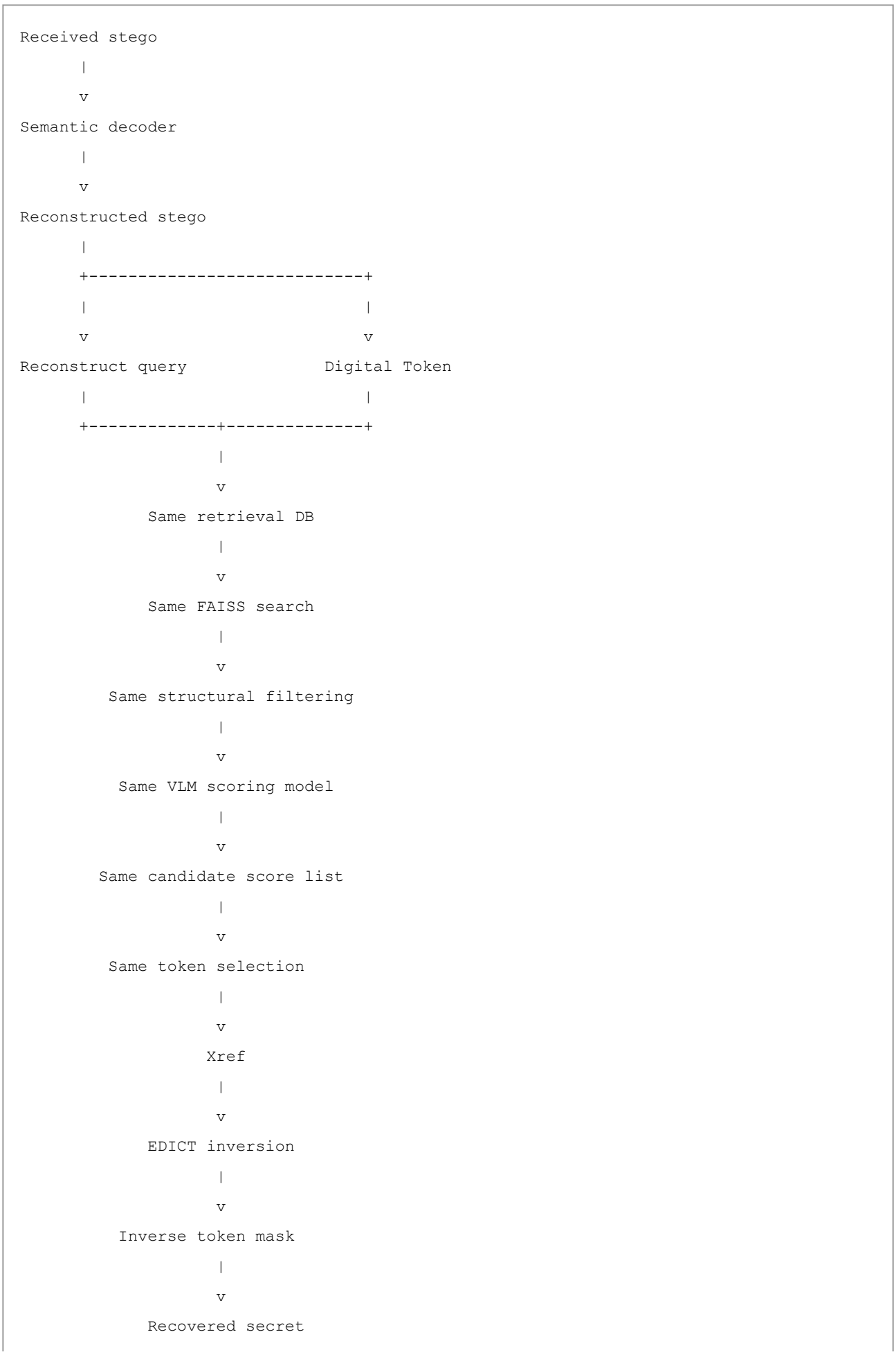


29. Proposed Receiver Architecture

The receiver must obtain the **same reference image** required for inversion.

This is one of the most important engineering constraints.

Proposed flow:


Received stego
Semantic decoder
Reconstructed stego
Reconstruct query
Digital Token
Same retrieval DB
Same FAISS search
Same structural filtering
Same VLM scoring model
Same candidate score list
Same token selection
Xref
EDICT inversion
Inverse token mask
Recovered secret

The database and retrieval configuration therefore need to be shared between legitimate transmitter and receiver.

---

30. Retrieval Database

The proposed system requires an indexed image corpus.

An initial manageable subset can be used, for example:

```
50,000–100,000 images
```

The database can later be scaled if required.

For each image we can store:

```
Image ID
Image
CLIP / VLM image embedding
Semantic metadata / caption
Structural representation
Optional suitability features
```

Example:

```
image_000124
      |
      +-- image
      +-- CLIP embedding
      +-- caption
      +-- structural representation
      +-- optional texture/entropy features
```

The database is created offline.

This is important for the latency argument because the expensive image-embedding computation happens during indexing rather than for every secret image.

---

31. Retrieval Query

The secret image is converted into a retrieval query.

The first version can use:

```
Secret image
     ↓
Semantic extraction
     ↓
Kpub
     ↓
Text / multimodal embedding
     ↓
FAISS search
```

This produces:

```
Top-K semantically similar images
```

However:

> ***Semantic similarity alone is not enough.***

The structural feature `Xfeat` must also influence the final candidate selection.

---

32. Structure-Aware Retrieval

We can define a candidate score:

```
Score(I) =

    α × SemanticSimilarity(I, Kpub)

  + β × StructuralSimilarity(I, Xfeat)
```

where:

- `I` = candidate image
- `Kpub` = public semantic key
- `Xfeat` = structural feature
- α = semantic weight
- β = structural weight

A more advanced version can include:

```
Score(I) =

    α Ssemantic
  + β Sstructural
  + γ Ssteganographic
```

where `Ssteganographic` estimates whether the image is a good reference for the downstream hiding mechanism.

This third term should initially remain optional until the simpler system is validated.

---

33. VLM / Agentic Candidate Evaluation

The VLM is not responsible for searching the entire database.

FAISS performs the large-scale search.

The VLM only evaluates a small candidate set.

```
Database
   |
   v
FAISS
   |
   v
Top 50 / Top 100
   |
   v
VLM
   |
   +--> semantic compatibility
   +--> structural compatibility
   +--> visual compatibility
   +--> downstream suitability
   |
   v
Candidate score
```

This gives us a practical compromise:

```
FAISS → speed
VLM   → intelligence
Token → deterministic selection
```

---

34. How the VLM Should Be Used

The VLM should produce a deterministic compatibility score rather than an open-ended generated answer.

Conceptually:

```
Candidate image
      +
Kpub
      +
Xfeat
      ↓
VLM
      ↓
Compatibility score
```

For example:

```
Candidate 1 → 0.82
Candidate 2 → 0.91
Candidate 3 → 0.76
...
```

The candidate pool is sorted by score.

This is preferable to:

```
"Which image do you like?"
```

because a numeric scoring formulation is:

- easier to reproduce,
- easier to benchmark,
- easier to log,

- easier to debug,
- easier to run at both transmitter and receiver.

---

35. Digital Token in RACS

The digital token is retained.

However, its reference-selection role changes.

Original:

```
Token → diffusion noise seed
```

RACS:

```
Token → deterministic candidate selection
```

For example:

```
VLM-ranked candidates:

C1
C2
C3
...
CK

Token
  ↓
deterministic mapping
  ↓
selected candidate
```

Conceptually:

```
index = H(token) mod K
```

where `H` is a deterministic hash-like mapping.

The token should select **only from a high-quality candidate pool**.

---

36. Why Token Selection Matters

If we always choose the highest-ranked candidate, retrieval becomes highly deterministic.

That is useful for reproducibility but reduces token-controlled diversity.

The token can provide controlled selection:

```
Query
  ↓
Top 100 candidates
  ↓
VLM ranking
  ↓
valid high-quality pool

Token A → candidate 17
Token B → candidate 83
Token C → candidate 41
```

All selected candidates should remain above a minimum compatibility threshold.

This creates a possible additional security interpretation:

```
Attacker knows:
    query
    database
    retrieval algorithm

Attacker does not know:
    token

Attacker must determine:
    selected candidate
```

This should be experimentally evaluated rather than automatically claimed as a security proof.

---

37. Important Constraint: Token Selection Must Not Destroy Quality

Bad design:

```
Token
  ↓
Random database image
```

Good design:

```
Semantic query
      ↓
Top-K candidates
      ↓
Structural filtering
      ↓
VLM reranking
      ↓
High-quality candidate pool
      ↓
Token-controlled deterministic selection
```

This preserves both:

- retrieval relevance,
- token-controlled diversity/security.

---

38. Optional Lightweight Editing

Pure retrieval may not be structurally compatible enough with the secret image.

Therefore RACS can include an optional lightweight editing stage.

```
Retrieved image
      +
Secret structural feature
      ↓
Light image-to-image editing
      ↓
Structurally aligned reference
```

Possible directions include:

- SDEdit-style partial diffusion,
- lightweight image-to-image adjustment,
- structure-guided editing.

The key principle is:

> ***The editing stage must remain substantially cheaper than full reference generation.***

Otherwise the efficiency motivation disappears.

---

39. Retrieve-Only vs Intelligent Retrieval

The project should not immediately assume that the most complex architecture is the best.

We should explicitly evaluate:

```
RACS-Base
FAISS only

        vs

RACS-Agent
FAISS + structural filtering + VLM reranking

        vs

RACS-AgentEdit
FAISS + VLM + lightweight editing
```

This gives us an important scientific comparison.

---

40. Optional Hybrid Retrieve-or-Generate Architecture

If retrieval fails for difficult images, a hybrid system can be introduced.

```
Secret image
      ↓
FAISS retrieval
      ↓
VLM evaluates best candidate
      ↓
Quality threshold?
     / \
   YES  NO
   |     |
   v     v
Retrieve  Diffusion generation
fast      expensive fallback
```

This is a strong possible final architecture because it does not completely discard the original generative mechanism.

The system generates only when retrieval is insufficient.

However, this should be treated as a later experiment, not mixed into the first baseline comparison.

---

41. The Core Research Comparison

We should compare the systems in controlled stages.

Baseline

```
AgentSemSteCom
```

Reference:

```
Diffusion-generated Xref
```

Proposed 1

```
RACS-Base
```

Reference:

```
FAISS-retrieved Xref
```

Proposed 2

```
RACS-Agent
```

Reference:

```
FAISS
+
structure-aware filtering
+
VLM reranking
+
token selection
```

Proposed 3

```
RACS-AgentEdit
```

Reference:

```
FAISS
+
VLM
+
lightweight editing
```

Optional Proposed 4

```
RACS-Hybrid
```

Reference:

```
Retrieve when possible
Generate when necessary
```

Everything else should remain as similar as possible.

---

42. Baseline vs RACS

Component	Existing AgentSemSteCom	Proposed RACS
Secret image	Same	Same
Semantic extraction	Kpub	Same
Structural feature	Xfeat	Same
Reference acquisition	Diffusion generation	Retrieval
Reference source	Generated	Indexed real image
Candidate database	Not required	Required
Vector embeddings	Not required	CLIP/VLM
Search	Not required	FAISS
Candidate evaluation	Diffusion conditioning	Structural + VLM
Token	Yes	Yes
Token role	Generation seed + latent perturbation	Candidate selection + latent perturbation

Component	Existing AgentSemSteCom	Proposed RACS
Structural conditioning	ControlNet	Retrieval/reranking + optional editing
Hiding core	EDICT	Same initially
Token perturbation	Yes	Yes
Semantic communication	Same	Same
Channel	Same	Same
Receiver	EDICT inversion	Same inversion logic
Main change	Generation	Reference acquisition

---

43. Core Research Hypotheses

H1 — Efficiency hypothesis

Retrieval-based reference acquisition reduces:

- reference-generation latency,
- GPU computation,
- memory requirements,
- computational cost,

compared with diffusion-based reference generation.

H2 — Quality hypothesis

A sufficiently relevant retrieved reference image, especially after intelligent reranking and optional lightweight editing, can preserve acceptable:

- PSNR,
- SSIM,
- MSE,
- LPIPS

relative to AgentSemSteCom.

H3 — Intelligent retrieval hypothesis

VLM/agentic reranking will improve downstream reconstruction quality compared with pure semantic retrieval because it can consider structural compatibility.

H4 — Security hypothesis

Token-controlled selection can preserve a meaningful legitimate-vs-wrong-token security gap.

H5 — Robustness hypothesis

RACS should remain usable across:

```
5, 10, 15, 20 dB
```

although its performance may differ from the original system.

---

44. The Most Important Scientific Question

The project should NOT be framed as:

> *“RAG is faster than diffusion.”*

That is too shallow.

The stronger research question is:

> ***Can intelligent retrieval-based reference-image acquisition provide sufficiently compatible semantic and structural conditioning for reversible coverless steganography, while substantially reducing the computational cost of reference generation?***

This gives us a real quality-vs-efficiency trade-off to investigate.

---

45. Expected Advantages

If successful, RACS could provide:

1. Lower latency

Retrieval from a precomputed FAISS index can be much faster than running a full diffusion generation process.

2. Lower computational cost

Image embeddings are precomputed offline.

Online processing mainly performs:

```
query embedding
+
FAISS search
+
small candidate reranking
+
optional lightweight editing
```

3. Real-image references

The reference image comes from an existing corpus rather than being generated from random noise.

4. Intelligent reference selection

VLM reranking can account for properties that a simple cosine-similarity search may miss.

5. Scalable retrieval

The reference database can grow without changing the core steganographic mechanism.

6. New security design space

Token-controlled candidate selection creates a discrete reference-selection mechanism.

---

46. Main Risks

Risk 1 — Retrieval quality is insufficient

The nearest semantic image may not be structurally compatible.

Consequence:

```
Poor reference
     ↓
Poor EDICT conditioning
     ↓
Lower PSNR / SSIM
```

Mitigation

Use:

- structural-aware reranking,
- Top-K candidate search,
- VLM evaluation,
- lightweight editing,
- retrieve-or-generate fallback.

Risk 2 — VLM reranking removes the efficiency advantage

If candidate evaluation becomes too expensive:

```
Fast retrieval
      +
Very expensive VLM
      ↓
Small overall speedup
```

Mitigation

Benchmark separately:

```
FAISS only
FAISS + VLM
FAISS + VLM + editing
Diffusion generation
```

Use a small/frozen VLM and restrict evaluation to a small Top-K.

Risk 3 — VLM is nondeterministic

If the transmitter chooses candidate A and the receiver chooses candidate B:

```
Xref_tx ≠ Xref_rx
```

EDICT recovery may fail.

Mitigation

Use:

- same model,
- same model version,
- deterministic inference,
- fixed preprocessing,
- fixed candidate ordering,
- deterministic scoring,
- deterministic token selection.

Risk 4 — Lightweight editing becomes too expensive

If editing uses too many diffusion steps, the efficiency advantage disappears.

Mitigation

Benchmark:

```
retrieval only
retrieval + light editing
full generation
```

Risk 5 — Reference mismatch at receiver

The receiver must use exactly the same reference.

Mitigation

Ensure:

- shared database version,
- identical FAISS index,
- identical embedding model,
- identical preprocessing,
- identical ranking procedure,
- identical VLM,
- deterministic token selection,
- identical editing parameters.

---

47. Optional Research Contribution — Steganographic Suitability Retrieval

Normal CLIP retrieval asks:

> *“Which image is semantically similar?”*

Our system could ask:

> *“Which image is semantically similar* ***and suitable for the downstream steganography mechanism****?”*

A learned reranking model could consider:

- texture,
- local entropy,
- high-frequency content,
- reconstruction quality,
- steganalysis detectability,
- downstream EDICT recovery quality.

This turns retrieval itself into a steganography-aware component.

This should be considered only after the basic retrieval system works.

---

48. Optional Research Contribution — Agentic Retrieve-or-Generate

Pure retrieval can fail for unusual images.

Therefore:

```
Secret image
      ↓
Retrieval
      ↓
VLM evaluates candidate
      ↓
Best candidate quality?
      |
   +--+--+
   |     |
  YES    NO
   |     |
   v     v
Retrieve Generation
fast     slow
path     fallback
```

This produces an adaptive architecture.

The system uses generation only when retrieval is not good enough.

---

49. Optional Research Contribution — Security

Formalization

The original token is associated with continuous diffusion noise.

Our retrieval design introduces a discrete selection problem.

If the token selects one item from a candidate pool of size `K`, a simple starting point is:

```
P(correct reference | no token) ≈ 1/K
```

and the corresponding selection uncertainty can be expressed approximately as:

```
log2(K)
```

bits.

This must be treated as a research hypothesis/model rather than automatically claimed as a complete security proof.

The final security analysis should account for:

- database knowledge,
- candidate ranking knowledge,
- token entropy,
- candidate filtering,
- attacker capabilities,
- side information.

---

50. Optional Research Contribution — Structure-Aware Multimodal Retrieval

Instead of:

```
CLIP semantic similarity only
```

we can use:

```
semantic similarity
+
structural similarity
```

This directly addresses one of the main weaknesses of naïve retrieval.

The original paper itself shows that the choice of structural feature affects reconstruction quality.

Therefore structural compatibility is directly relevant to downstream reconstruction.

---

51. Evaluation Framework

Our final benchmark should compare:

```
AgentSemSteCom

        VS

RACS Retrieval-only

        VS

RACS + VLM/Agent Reranking

        VS

RACS + VLM + Light Editing

        VS

RACS Hybrid Retrieve-or-Generate
```

The hybrid system should be included only if implemented.

This gives us a much stronger experimental story than comparing only two systems.

---

52. Quality Metrics

For every system calculate:

Metric Desired direction

Metric	Desired direction
PSNR	Higher
SSIM	Higher
MSE	Lower
LPIPS	Lower

Evaluate at:

```
SNR = 5 dB
SNR = 10 dB
SNR = 15 dB
SNR = 20 dB
```

and across the relevant image classes.

---

53. Efficiency Metrics

This is the major new evaluation dimension.

Reference acquisition latency

```
time to obtain Xref
```

Compare:

```
Diffusion generation
vs
FAISS retrieval
vs
FAISS + VLM
vs
FAISS + VLM + editing
```

VLM reranking latency

Measure separately:

```
FAISS search time
+
candidate filtering time
+
VLM scoring time
```

This tells us whether intelligent retrieval remains computationally worthwhile.

End-to-end latency

```
semantic extraction
+
reference acquisition
+
steganography
+
communication
+
recovery
```

GPU memory

Measure peak GPU memory for:

- reference generation,
- retrieval,
- VLM reranking,
- editing,
- full pipeline.

Compute

Where feasible:

- FLOPs estimate,
- diffusion steps,
- number of model forward passes,
- retrieval operations.

Optional:

- energy consumption,

- CPU utilization,
- GPU utilization.

---

54. Security Metrics

At minimum reproduce:

```
Correct token
Wrong token
No token
```

Measure:

- recovered secret PSNR,
- recovered secret SSIM,
- MSE,
- LPIPS.

The baseline paper reports:

```
Correct token: 25.37 dB PSNR
Wrong token:   18.91 dB PSNR
```

Our RACS security experiment should use the same conceptual setup.

For RACS, we should additionally investigate whether an attacker who knows the database and retrieval method can infer the selected reference without the token.

---

55. Steganalysis

If feasible, evaluate whether an attacker can distinguish:

```
Normal images
vs
Stego images
```

using steganalysis models such as:

- XuNet
- SRNet

Metrics can include:

```
Detection accuracy
AUC
False positive rate
False negative rate
```

Lower detectability is preferable.

This should be included if implementation resources permit.

---

56. Retrieval Ablation Experiments

We should isolate the effect of each design choice.

Experiment A

```
Pure semantic retrieval
```

Experiment B

```
Semantic + structural reranking
```

Experiment C

```
FAISS + VLM reranking
```

Experiment D

```
FAISS + VLM + token selection
```

Experiment E

```
Retrieval + light editing
```

Experiment F

```
Hybrid retrieve-or-generate
```

This lets us determine exactly where improvements/degradations come from.

---

57. Top-K Ablation

The retrieval candidate pool size should be varied.

For example:

```
K = 10
K = 50
K = 100
K = 500
```

The exact values can be finalized during implementation based on compute availability.

Measure:

```
K
 ↓
retrieval quality
 ↓
VLM ranking quality
 ↓
reconstruction quality
 ↓
security
 ↓
latency
```

The goal is to find the smallest candidate pool that provides strong reference quality without unnecessary VLM computation.

---

58. Expected Comparison Table

The final project should produce a table approximately like:

System	PSNR ↑	SSIM ↑	MSE ↓	LPIPS ↓	Ref. latency ↓	VLM latency ↓	GPU memory ↓	Security gap ↑
AgentSemSteCom	baseline	baseline	baseline	baseline	baseline	—	baseline	baseline
RACS Retrieval	?	?	?	?	?	—	?	?
RACS + VLM	?	?	?	?	?	?	?	?
RACS + VLM + Editing	?	?	?	?	?	?	?	?
RACS Hybrid	?	?	?	?	?	?	?	?

The question marks are intentional.

We should never invent results before running the experiments.

---

59. What Would Count as a Successful Result?

The project does **not** require RACS to beat the original system on every metric.

A scientifically meaningful result could be:

```
PSNR: slightly lower
SSIM: similar
LPIPS: similar
Security: similar
Latency: dramatically lower
GPU memory: lower
```

That would demonstrate a useful quality-efficiency trade-off.

Another possible result:

```
Retrieval-only:
fast but quality drops significantly

Retrieval + VLM:
quality improves

Retrieval + editing:
quality recovers further

Hybrid:
generation used only for difficult cases
```

That could be an even stronger result.

---

60. What Would Count as a Negative Result?

A negative result is also useful if properly analyzed.

For example:

```
Retrieval is 8× faster

BUT

PSNR decreases by 7 dB
SSIM decreases substantially
EDICT recovery becomes unstable
```

Then the conclusion may be:

> ***Pure retrieval cannot provide sufficiently compatible reference images for the existing reversible diffusion mechanism.***

This would motivate:

```
structure-aware retrieval
or
VLM reranking
or
light editing
or
retrieve-or-generate
```

The project should therefore be designed so that every experiment teaches us something.

---

61. Critical Fairness Rule for the Comparison

When comparing AgentSemSteCom and RACS, we should keep as many variables identical as possible.

Ideally:

```
Same dataset
Same secret images
Same preprocessing
Same Kpub extraction
Same Xfeat extraction
Same EDICT configuration
Same token perturbation
Same semantic communication model
Same channel
Same SNR
Same evaluation metrics
Same evaluation code
```

Only the **reference acquisition method** should change in the first comparison.

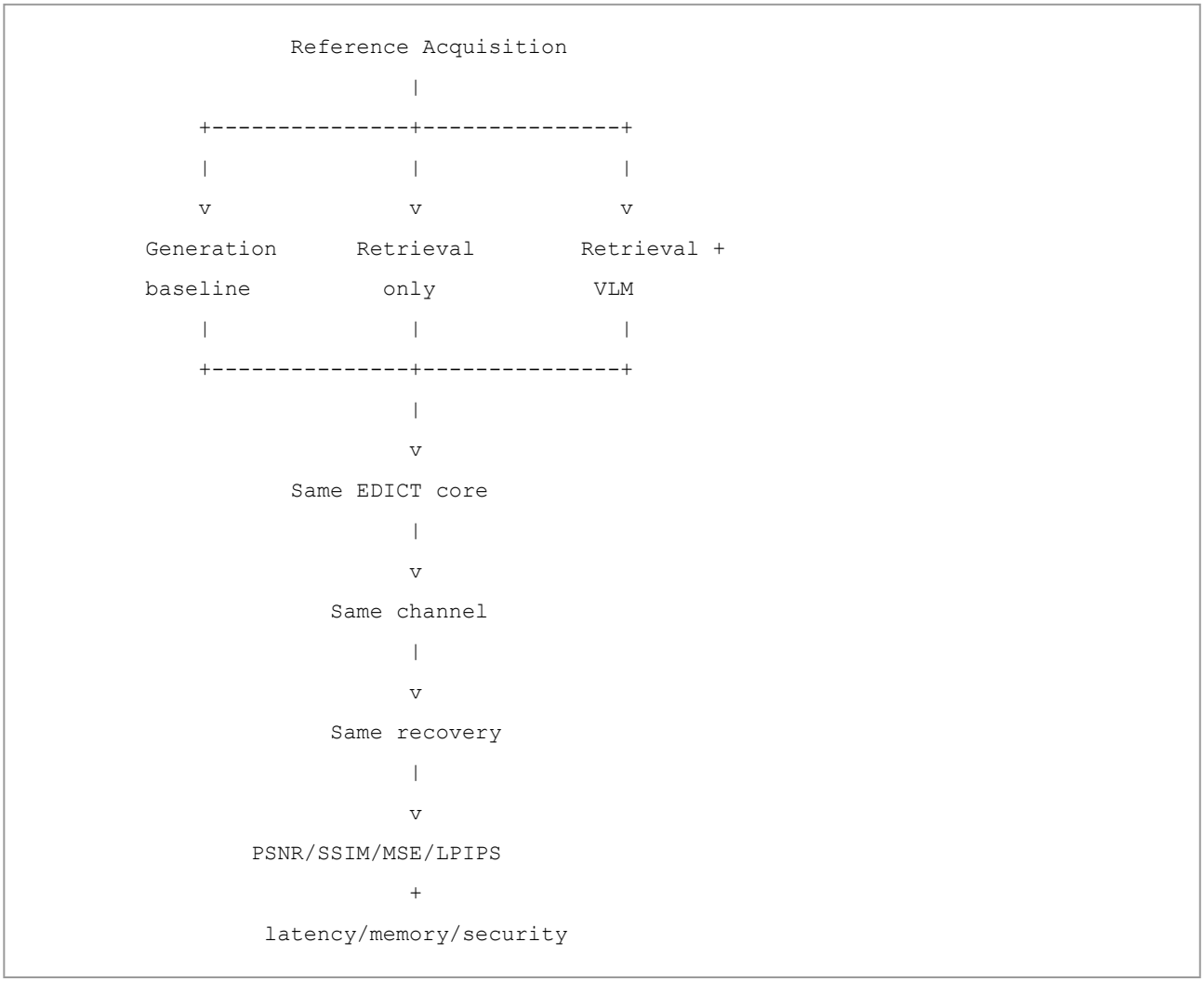
This is extremely important.

Otherwise we cannot confidently attribute an improvement or degradation to retrieval.

---

62. Recommended Experimental Matrix

For each test image:



Then extend:

```
Image class
×
SNR
×
reference method
×
token condition
```

---

63. Proposed End-to-End RACS Flow

Transmitter

1. Input secret image Xs

2. Extract public semantic key Kpub

3. Extract implicit structural feature Xfeat

4. Construct retrieval query

5. Convert query to embedding

6. Search FAISS image index

7. Retrieve Top-K candidates

8. Apply structural filtering

9. Score candidates using deterministic VLM/agentic model

10. Construct high-quality candidate pool

11. Use digital token to deterministically select
    a candidate from the valid pool

12. Optionally perform lightweight structure-aware editing

13. Produce reference image Xref

14. Run the existing EDICT-based coverless
    steganography mechanism

15. Apply digital-token latent perturbation

16. Produce stego image Xstego

17. Encode with semantic communication model

18. Transmit through wireless channel

Receiver

```
1. Receive channel-corrupted semantic representation

2. Decode semantic representation

3. Recover reconstructed stego image

4. Reconstruct the same retrieval query

5. Search the same shared retrieval database

6. Apply the same structural filtering

7. Apply the same deterministic VLM scoring

8. Use the correct digital token to reproduce
   the selected reference image

9. Apply the same lightweight editing procedure
   if used

10. Perform EDICT inversion

11. Apply inverse token perturbation

12. Recover secret image

13. Compute quality metrics
```

---

64. The Most Important Architectural Constraint

The transmitter and receiver must agree on:

```
Database version
Embedding model
Preprocessing
FAISS index
Ranking function
VLM model/version
VLM inference configuration
Top-K value
Token mapping
Editing parameters
EDICT parameters
```

Otherwise:

```
Xref_tx ≠ Xref_rx
```

and the reversible recovery mechanism can fail.

This reproducibility problem should be treated as a first-class design requirement.

---

65. Project Components

The implementation should eventually be divided into independent modules.

```
RACS/
│
├── data/
│   ├── secret_images/
│   ├── retrieval_images/
│   └── metadata/
│
├── retrieval/
│   ├── embedding/
│   ├── faiss_index/
│   ├── structural_filter/
│   ├── vlm_ranking/
│   └── token_selection/
│
├── semantic/
│   ├── semantic_extraction/
│   └── structure_extraction/
│
├── reference/
│   ├── retrieval/
│   └── lightweight_editing/
│
├── steganography/
│   ├── vae/
│   ├── edict/
│   └── token_perturbation/
│
├── communication/
│   ├── awgn/
│   ├── swinjscc/
│   └── digital_semcom/
│
├── security/
│   ├── wrong_token/
│   ├── no_token/
│   └── steganalysis/
│
├── evaluation/
│   ├── psnr.py
│   ├── ssim.py
│   ├── mse.py
│   ├── lpips.py
│   ├── latency.py
│   └── memory.py
```

```
│
└── experiments/
    ├── baseline/
    ├── retrieval/
    ├── vlm_reranking/
    ├── editing/
    ├── ablation/
    └── results/
```

This is an architectural target, not yet the implementation plan.

---

66. What We Are NOT Changing Initially

To keep the research comparison scientifically clean, the first version should NOT simultaneously replace everything.

We should initially keep:

```
Semantic extraction       → same

Implicit feature          → same

EDICT                     → same

Token perturbation        → same

Semantic communication    → same

Channel                   → same

Recovery                  → same

Evaluation metrics        → same
```

The primary changed module is:

```
Reference acquisition
```

This makes the experiment interpretable.

---

67. Step-Efficient Denoising — Secondary Research Direction

The original paper explicitly identifies **step-efficient denoising for coverless steganography** as future work.

Therefore, after the core RACS comparison is stable, we can investigate:

```
Original EDICT denoising
        ↓
Reduced-step / efficient denoising
```

The objective is:

```
Fewer denoising operations
        ↓
Lower latency
        ↓
Lower computation
        ↓
Acceptable reconstruction quality
```

This should be evaluated as a separate controlled modification.

The experimental order should therefore be:

```
AgentSemSteCom
      ↓
RACS retrieval
      ↓
RACS + intelligent retrieval
      ↓
RACS + optional editing
      ↓
RACS + efficient denoising
```

This prevents two different optimizations from being mixed together before we understand their individual effects.

---

68. What We May Add After the Core System Works

After the baseline and retrieval versions are working, we can add incrementally:

1. Structure-aware retrieval
2. VLM/agentic reranking
3. Learned steganographic suitability scoring
4. Lightweight editing
5. Retrieve-or-generate agent
6. Formal token/index security analysis
7. Larger retrieval database
8. Advanced steganalysis
9. Step-efficient denoising
10. Model compression

These should be added incrementally rather than all at once.

---

69. Research Contributions We Can Aim For

The strongest final version could claim contributions along these lines:

Contribution 1

A retrieval-based reference-image acquisition framework for coverless semantic steganography.

Contribution 2

A multimodal retrieval/reranking mechanism that considers both semantic relevance and structural compatibility.

Contribution 3

A deterministic token-controlled reference selection mechanism.

Contribution 4

An optional lightweight editing mechanism for improving reference compatibility without performing full reference generation.

Contribution 5

An adaptive retrieve-or-generate strategy, if implemented and validated.

Contribution 6

A comprehensive quality-security-efficiency evaluation demonstrating the trade-off between diffusion generation and retrieval.

Contribution 7

A step-efficient denoising study inspired by the original paper's explicit future work.

The final paper should only claim contributions that are actually implemented and experimentally validated.

---

70. Project Success Criteria

At the end of the project, we should be able to answer all of these.

Quality

- Does retrieval preserve reconstruction quality?
- How much does PSNR change?
- How much does SSIM change?
- How does MSE change?
- How does LPIPS change?

Efficiency

- How much faster is retrieval?
- How much time does VLM reranking add?
- How much GPU memory is saved?
- How much computation is saved?
- What is the cost of optional editing?

Security

- Does the correct token still enable recovery?
- Does a wrong token fail?

- What happens without a token?
- Does candidate selection create a meaningful security gap?
- Does the retrieval database introduce new attack surfaces?

Robustness

- How does performance change from 5 to 20 dB?
- Which image classes benefit?
- Which image classes fail?

Retrieval

- How does Top-K affect quality?
- Does structural reranking improve recovery?
- Does VLM reranking improve over pure CLIP retrieval?
- Does token-controlled selection preserve quality?

Architecture

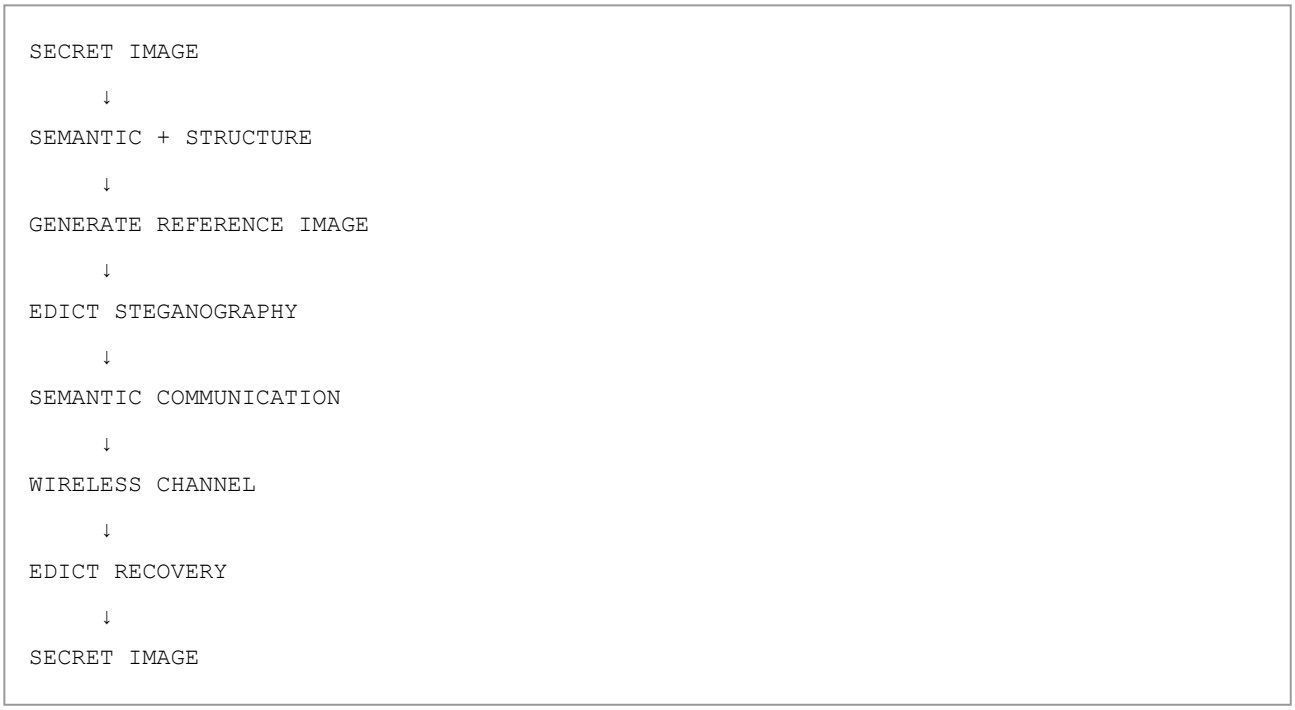
- When should the system retrieve?
- When should it use VLM reranking?
- When should it edit?
- When should it fall back to generation?

---

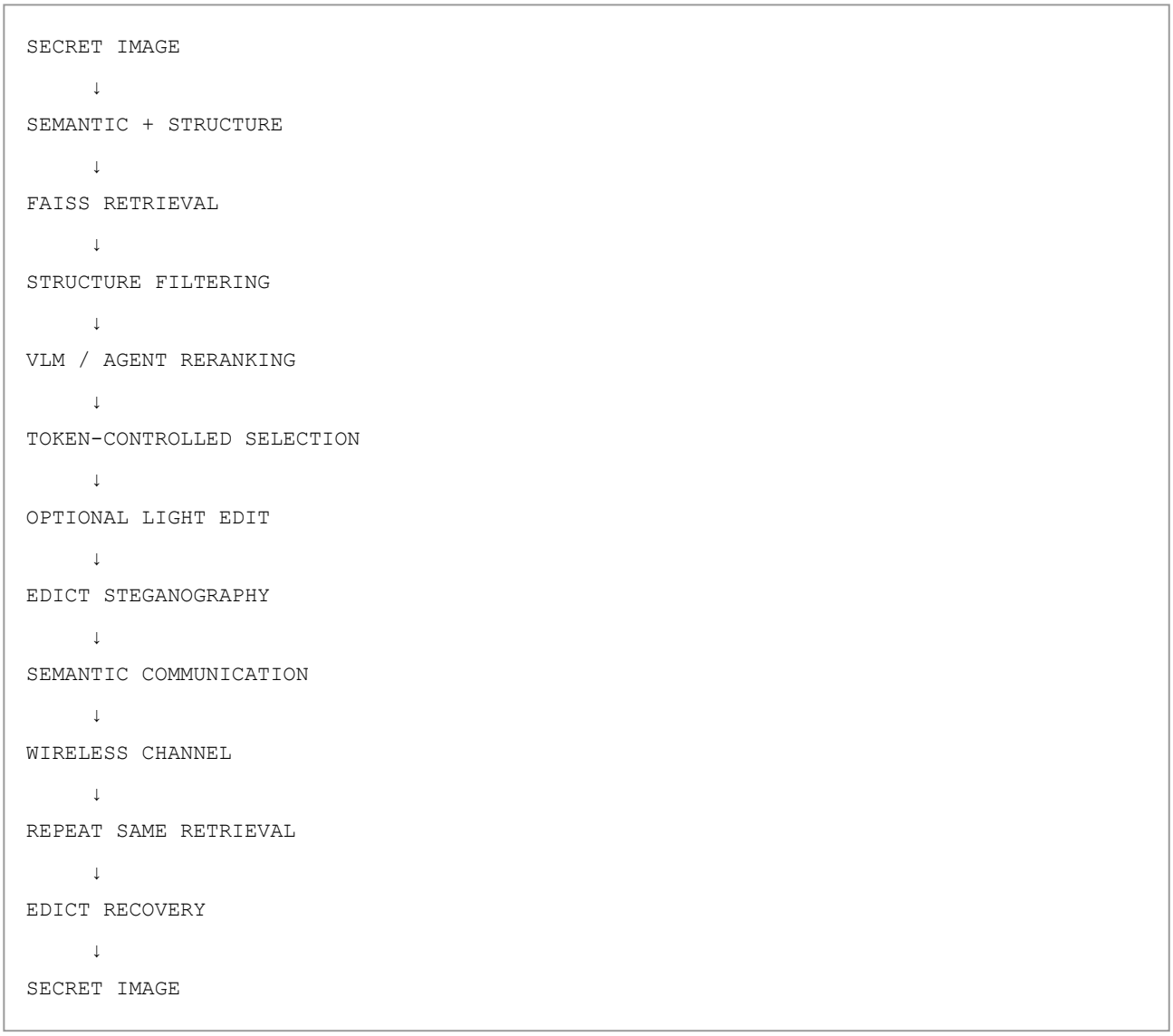
71. Final Mental Model

The easiest way to remember the entire project is:

Existing system


SECRET IMAGE
↓
SEMANTIC + STRUCTURE
↓
GENERATE REFERENCE IMAGE
↓
EDICT STEGANOGRAPHY
↓
SEMANTIC COMMUNICATION
↓
WIRELESS CHANNEL
↓
EDICT RECOVERY
↓
SECRET IMAGE

Our system



The research question

```
Can we replace:

        expensive reference generation

with:

        efficient intelligent retrieval

without sacrificing too much:

        quality
        security
        robustness
        reversibility

while gaining:

        lower latency
        lower computation
        lower memory
        better suitability for resource-constrained systems
```

---

72. One-Sentence Project Definition

> ***RACS is a retrieval-augmented coverless semantic steganography framework that replaces the computationally expensive diffusion-based reference-image generation stage of AgentSemSteCom with deterministic multimodal retrieval, structure-aware filtering, and intelligent VLM/agentic reranking, with optional lightweight editing, while preserving the downstream EDICT-based hiding/recovery and semantic communication pipeline so that quality, security, robustness, and computational efficiency can be directly compared.***

---

73. What Comes Next

The next document should be the **Implementation Plan**, not another conceptual redesign.

It should be built directly from this overview and answer:

1. What exactly do we reproduce first from AgentSemSteCom?
2. What exact baseline metrics do we obtain ourselves?
3. What code/modules are required?
4. What models/checkpoints/datasets are needed?
5. What should be implemented first?
6. How do we build the retrieval database?
7. How do we implement CLIP + FAISS?
8. How do we implement structural filtering?
9. How do we implement deterministic VLM scoring?
10. How do we implement token-controlled candidate selection?
11. How do we implement lightweight editing?
12. How do we integrate RACS with EDICT?
13. How do we ensure transmitter/receiver reference consistency?
14. How do we run the exact same test set for both systems?
15. How do we log every metric?
16. How do we compare improvements/degradations?
17. What experiments and ablations must be run?
18. How do we benchmark the VLM overhead?
19. What should be done if retrieval quality is poor?
20. What is the fallback path?
21. How do we investigate step-efficient denoising separately?
22. How do we produce the final tables and figures?

The implementation plan should **not change the baseline or evaluation definitions casually**.

The purpose is to create a reproducible experiment where every improvement or degradation can be attributed to a specific architectural change.